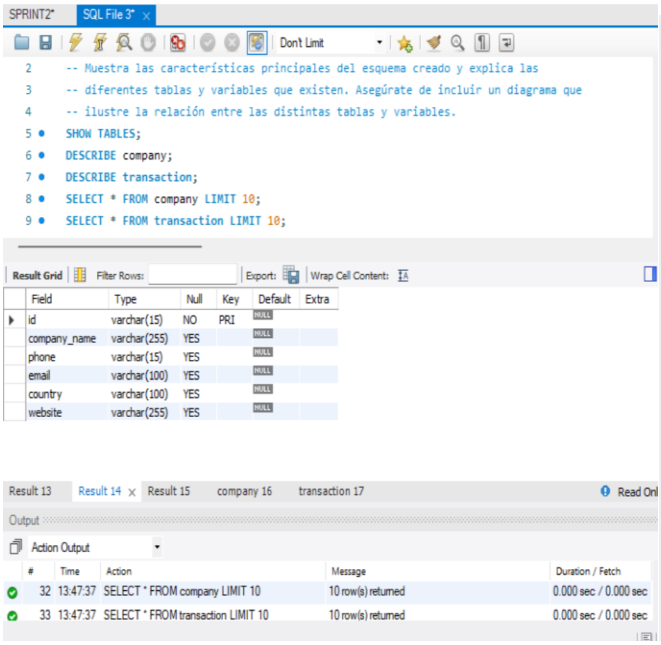
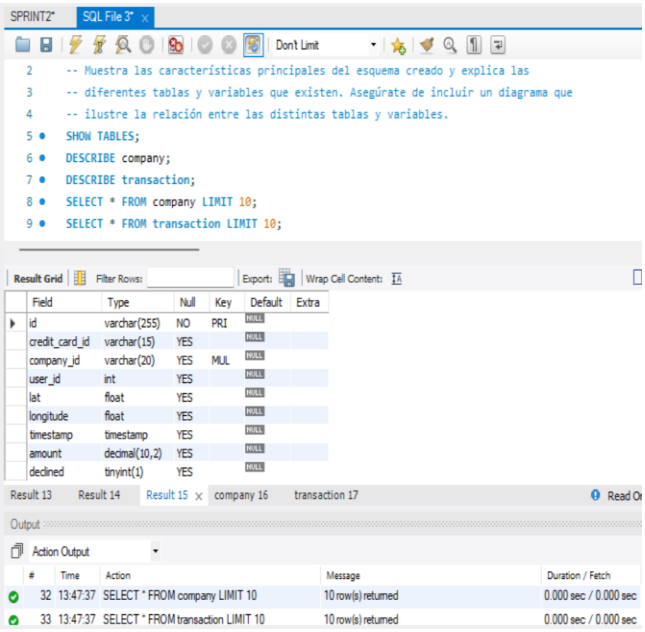
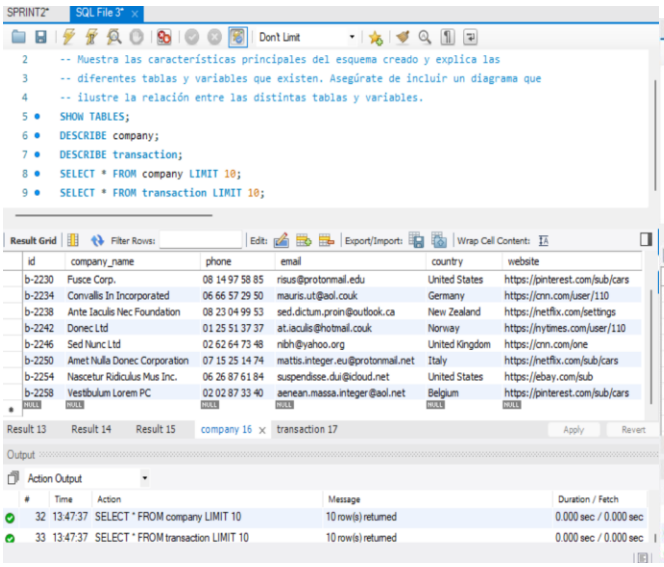
Nivell 1

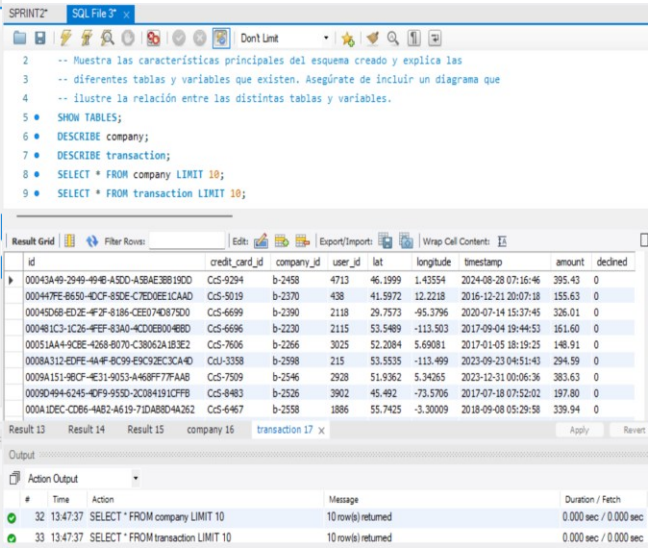
- **Ejercicio 1**

A partir de los documentos adjuntos (estructura_dades y dades_introduir), importa las dos tablas. Muestra las características principales del esquema creado y explica las diferentes tablas y variables que existen. Asegúrate de incluir un diagrama que ilustre la relación entre las diferentes tablas y variables.

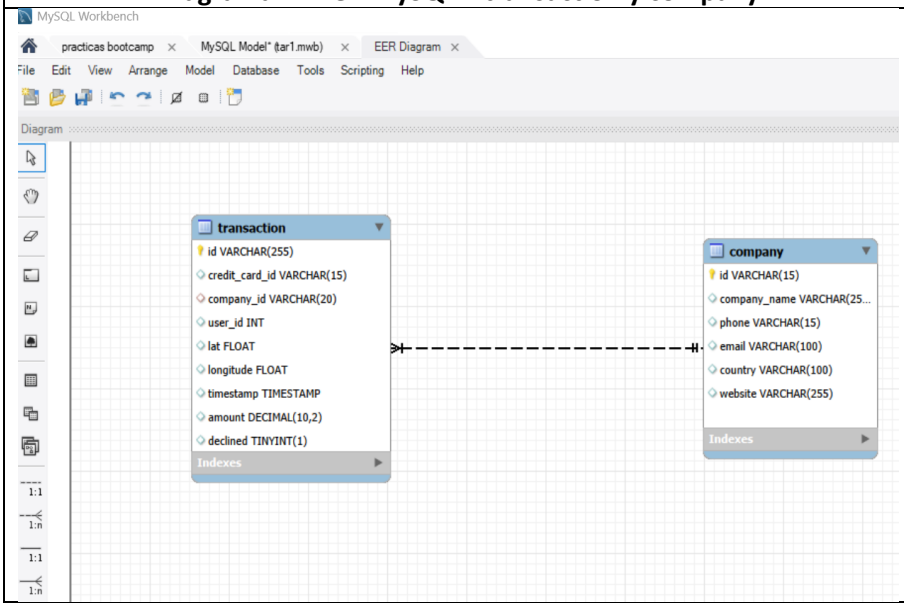








Comenzando el Sprint 2 se comenzó a la descarga y carga de la base de datos proporcionada, el cual por medio de los comandos de MySQL se comenzó con la observación, interpretación y análisis de las 2 tablas como ser: company y transaction (entender todos los elementos de construcción). A continuación, por medio de MySQL Model se observa el diagrama de relación de ambas tablas que adjunto se detalla sus características

Diagrama EER en MySQL – transaction y company	Características
	**Tipo de relación:** 1 a N (Uno a Muchos) **Claves** **Primary Key (PK):** company.id **Foreign Key (FK):** transaction.company_id **Caracteristicas de los datos** **VARCHAR(n):** texto de longitud variable. **INT:** números enteros. **FLOAT:** números decimales aproximados (coordenadas). **DECIMAL:** números exactos (dinero). **TIMESTAMP:** fecha y hora. **TINYINT(1):** booleano (0 o 1).

- Ejercicio 2: Utilizando JOIN, realizarás las siguientes consultas:

  Listado de los países que están generando ventas.

```
23
24 •  SELECT DISTINCT c.country
25    FROM company c
26    JOIN transaction t ON c.id = t.company_id
27    WHERE t.declined = 0;
28
29 •  SELECT COUNT(DISTINCT c.country) AS paisescon_fact
30    FROM company c
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country
Netherlands
Sweden
Ireland
United States
Belgium
Canada
Germany
Norway
France
Italy

Result 8

Output

Action Output

#	Time	Action	Message
23	11:32:57	SELECT company_name,COUNT(*) AS total_trans, CASE WHEN COUNT(*) > 400 THEN 'mas de 400' ELS...	100 row(s) returned
24	13:03:03	SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined = 0	15 row(s) returned

Desde cuántos países se generan las ventas.

```
SELECT COUNT(DISTINCT c.country) AS paisescon_fact
FROM company c
JOIN transaction t ON c.id = t.company_id
WHERE t.declined = 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

paisescon_fact
15

Result 9

Output

Action Output

#	Time	Action	Message
24	13:03:03	SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined = 0	15 row(s) returned
25	13:08:03	SELECT COUNT(DISTINCT c.country) AS paisescon_fact FROM company c JOIN transaction t ON c.id = t.compa...	1 row(s) returned

Identifica la empresa con el mayor promedio de ventas.

```
SELECT c.company_name,ROUND(AVG(t.amount) ,2)AS promedio
FROM company c
JOIN transaction t on c.id = t.company_id
WHERE t.declined = 0
group by c.company_name
ORDER BY promedio DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	promedio
Ac Fermentum Incorporated	284.91

Result 10

Output

Action Output

#	Time	Action	Message
25	13:08:03	SELECT COUNT(DISTINCT c.country) AS paisescon_fact FROM company c JOIN transaction t ON c.id = t.compa...	1 row(s) returned
26	13:10:10	SELECT c.company_name,ROUND(AVG(t.amount) ,2)AS promedio FROM company c JOIN transaction t on c.id = t...	1 row(s) returned

- Ejercicio 3: Utilizando solo subconsultas (sin usar JOIN):

Muestra todas las transacciones realizadas por empresas de Alemania.

```
SELECT *
FROM transaction t
WHERE company_id IN ( SELECT id
                      FROM company
                      WHERE country = 'Germany'
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0
00687139-48B2-4FFA-8E73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4669	2019-03-09 19:37:49	524.84	0
0074F4DD-32F1-4827-8758-55896314623A	CcS-8081	b-2222	3500	39.7016	-8.50325	2016-12-26 23:06:57	491.90	0
00AAB9CD-39D6-4DCB-8A1D-13BE73DC90A9	CcS-6797	b-2222	2216	55.7652	-3.76245	2021-04-25 03:06:59	167.15	0
00BE09D4-6920-47D8-ABE8-325E2269829D	CcS-4983	b-2222	402	38.708	-9.12993	2019-02-27 15:25:16	141.66	0
00DA0383-E048-4577-8ED1-3C56C258FF2F	CcS-9223	b-2222	4642	51.1742	10.2027	2019-03-21 11:47:34	325.62	0

transaction 11

Output

Action Output

#	Time	Action	Message
26	13:10:10	SELECT c.company_name,ROUND(AVG(t.amount) ,2)AS promedio FROM company c JOIN transaction t on c.id = t...	1 row(s) returned
27	13:13:28	SELECT * FROM transaction t WHERE company_id IN (SELECT id FROM company WHERE coun...	13291 row(s) returned

Lista las empresas que han realizado transacciones por un importe superior al promedio de todas las transacciones.

```
SELECT company_name
FROM company
WHERE id IN ( SELECT company_id
              FROM transaction
              WHERE amount > ( SELECT AVG (amount)
                               FROM transaction
              )
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name
Ac Fermentum Incorporated
Magna A Neque Industries
Fusce Corp.
Convallis In Incorporated
Ante Iaculis Nec Foundation
Donec Ltd
Sed Nunc Ltd
Amet Nulla Donec Corporation
Nascetur Ridiculus Mus Inc.
Vestibulum Lorem PC

company 12

Output

Action Output

#	Time	Action	Message
27	13:13:28	SELECT * FROM transaction t WHERE company_id IN (SELECT id FROM company WHERE coun...	13291 row(s) returned
28	13:14:26	SELECT company_name FROM company WHERE id IN (SELECT company_id FROM transaction WHE...	100 row(s) returned

Se eliminarán del sistema las empresas que no tienen transacciones registradas; entrega el listado de estas empresas.

```
SELECT *
FROM company
WHERE id NOT IN (
    SELECT DISTINCT company_id FROM transaction
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	company_name	phone	email	country	website
NULL	NULL	NULL	NULL	NULL	NULL

company 13

Output

Action Output

#	Time	Action	Message
28	13:14:26	SELECT company_name FROM company WHERE id IN (SELECT company_id FROM transaction WHE...	100 row(s) returned
29	13:16:20	SELECT * FROM company WHERE id NOT IN (SELECT DISTINCT company_id FROM transaction)	0 row(s) returned

Nivel 2

Ejercicio 1: Identifica los cinco días en los que se generó la mayor cantidad de ingresos en la empresa por ventas. Muestra la fecha de cada transacción junto con el total de las ventas.

```
SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS  suma_total
FROM transaction t
WHERE declined = 0
GROUP BY  DATE(timestamp)
ORDER BY SUM(amount) DESC
limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

fecha	suma_total
2022-12-13	14337.44
2019-11-18	13591.32
2023-02-20	13332.59
2017-12-20	13318.43
2019-03-18	12680.95

Result 15

Output

Action Output

#	Time	Action	Message
32	13:31:50	SELECT DATE(timestamp) AS fecha, SUM(amount) AS suma_total FROM transaction t WHERE declined = 0 GRO...	5 row(s) returned
33	13:32:57	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS suma_total FROM transaction t WHERE declin...	5 row(s) returned

Ejercicio 2: ¿Cuál es el promedio de ventas por país? Presenta los resultados ordenados de mayor a menor promedio.

```
SELECT c.country, ROUND(AVG(amount),2) AS promedio
FROM    company c
JOIN transaction t ON c.id = t.company_id
WHERE declined =0
GROUP BY country
ORDER BY AVG(amount) DESC;

-- Ejercicio 3
-- En su empresa se propone un nuevo proyecto para lanzar algunas campañas publicitarias para
-- competir en la empresa "Non Institute". Para ello, le solicitan el listado de todas las
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	promedio
Australia	265.54
United States	264.42
Belgium	260.97
Germany	260.83
Ireland	260.39
Spain	260.28
France	259.91
New Zealand	259.59
Norway	259.14
Netherlands	258.34

Result 16

Output

Action Output

#	Time	Action	Message
33	13:32:57	SELECT DATE(timestamp) AS fecha, ROUND(SUM(amount),2) AS suma_total FROM transaction t WHERE declin...	5 row(s) returned
34	13:34:03	SELECT c.country, ROUND(AVG(amount),2) AS promedio FROM company c JOIN transaction t ON c.id = t.compa...	15 row(s) returned

Ejercicio 3: En tu empresa se plantea un nuevo proyecto para lanzar algunas campañas publicitarias con el objetivo de competir contra la compañía "Non Institute". Para ello, te solicitan la lista de todas las transacciones realizadas por empresas que están ubicadas en el mismo país que dicha compañía.

Muestra el listado aplicando JOIN y subconsultas.

```
SELECT c.country, t.timestamp,t.amount
FROM company c
JOIN transaction t ON c.id = t.company_id
WHERE c.country = ( SELECT country
                    FROM company
                    WHERE company_name = 'Non Institute'
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

country	timestamp	amount
United Kingdom	2015-07-30 12:12:42	486.44
United Kingdom	2017-10-26 22:08:26	414.06
United Kingdom	2018-01-27 13:44:36	103.73
United Kingdom	2016-12-19 11:53:45	219.28
United Kingdom	2017-01-10 21:09:29	326.87
United Kingdom	2020-05-24 01:17:29	155.72
United Kingdom	2023-03-17 16:36:27	305.35
United Kingdom	2021-12-02 23:06:02	339.58
United Kingdom	2021-12-30 08:40:24	172.93
United Kingdom	2018-11-11 11:28:49	114.77

Result 17

Output

Action Output

#	Time	Action	Message
34	13:34:03	SELECT c.country, ROUND(AVG(amount),2) AS promedio FROM company c JOIN transaction t ON c.id = t.compa...	15 row(s) returned
35	13:34:56	SELECT c.country, t.timestamp,t.amount FROM company c JOIN transaction t ON c.id = t.company_id WHERE c.c...	13776 row(s) returned

Muestra el listado utilizando únicamente subconsultas.

```
SELECT *
FROM transaction
WHERE company_id IN ( SELECT id
                      FROM company
                      WHERE country IN ( SELECT country
                                         FROM company
                                         WHERE company_name = 'Non Institute'
                                       )
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

country	timestamp	amount
United Kingdom	2015-07-30 12:12:42	486.44
United Kingdom	2017-10-26 22:08:26	414.06
United Kingdom	2018-01-27 13:44:36	103.73
United Kingdom	2016-12-19 11:53:45	219.28
United Kingdom	2017-01-10 21:09:29	326.87
United Kingdom	2020-05-24 01:17:29	155.72
United Kingdom	2023-03-17 16:36:27	305.35
United Kingdom	2021-12-02 23:06:02	339.58
United Kingdom	2021-12-30 08:40:24	172.93
United Kingdom	2018-11-11 11:28:49	114.77

Result 18

Output

Action Output

#	Time	Action	Message
35	13:34:56	SELECT c.country, t.timestamp,t.amount FROM company c JOIN transaction t ON c.id = t.company_id WHERE c.c...	13776 row(s) returned
36	13:36:11	SELECT c.country, t.timestamp,t.amount FROM company c JOIN transaction t ON c.id = t.company_id WHERE c.c...	13776 row(s) returned

Nivel 3

Ejercicio 1: Presenta el nombre, teléfono, país, fecha y amount de aquellas empresas que realizaron transacciones con un valor comprendido entre 350 y 400 euros y en alguna de estas fechas: 29 de abril de 2015, 20 de julio de 2018 y 13 de marzo de 2024. Ordena los resultados de mayor a menor cantidad.

```
SELECT  c.company_name, c.phone, c.country, t.amount, date(timestamp) AS fecha
FROM company c
JOIN transaction t ON c.id = t.company_id
WHERE amount BETWEEN 350 AND 400 AND DATE(timestamp) IN ('2015-04-29','2018-07-20','2024-03-13')
ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	country	amount	fecha
Aliquam PC	01 45 73 52 16	Germany	399.84	2024-03-13
Auctor Mauris Vel LLP	08 09 28 74 14	United States	399.51	2018-07-20
At Pede Corp.	06 14 48 33 15	Italy	390.69	2015-04-29
Aliquam PC	01 45 73 52 16	Germany	388.29	2024-03-13
Orci Adipiscing Limited	03 18 00 77 81	United Kingdom	373.71	2018-07-20
Fringilla LLC	08 29 15 93 57	New Zealand	367.62	2015-04-29
Pede Cum Ltd	07 62 26 48 38	Norway	356.87	2018-07-20
Auctor Mauris Vel LLP	08 09 28 74 14	United States	353.75	2024-03-13

Result 19

Output

Action Output

#	Time	Action	Message
36	13:36:11	SELECT c.country, t.timestamp,t.amount FROM company c JOIN transaction t ON c.id = t.company_id WHERE c.c...	13776 row(s) returned
37	13:37:29	SELECT c.company_name, c.phone, c.country, t.amount, date(timestamp) AS fecha FROM company c JOIN trans...	8 row(s) returned

Ejercicio 2: Necesitamos optimizar la asignación de los recursos en función de la capacidad operativa requerida. Por ello, te solicitan información sobre la cantidad de transacciones que realizan las empresas. Sin embargo, el departamento de recursos humanos es exigente y quiere un listado de las empresas donde se especifique si tienen más de 400 transacciones o menos.

```sql
SELECT company_name,COUNT(*) AS total_trans,
   CASE
      WHEN COUNT(*) > 400 THEN 'mas de 400'
      ELSE 'menos de 400'
   END AS categoria
FROM company c
JOIN transaction t ON c.id = t.company_id
WHERE declined =0
GROUP BY c.id, company_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	total_trans	categoria
Eget Tincidunt Dui Institute	1522	mas de 400
Non Justo Corp.	1476	mas de 400
Neque Tellus Imperdiet Corp.	422	mas de 400
Fusce Corp.	445	mas de 400
Mus Aenean Eget Foundation	1563	mas de 400
Aliquam Iaculis Lacus Corp.	422	mas de 400
Lorem Ipsum Dolor Corp.	1465	mas de 400
Amet Luctus Vulputate Foundation	410	mas de 400
Nulla Integer Vulputate Corp.	1447	mas de 400

Result 21

Output

Action Output

#	Time	Action	Message
41	13:38:41	SELECT company_name,COUNT(*) AS total_trans, CASE WHEN COUNT(*) > 400 THEN 'mas de 400' ELS...	100 row(s) returned
42	13:38:55	SELECT company_name,COUNT(*) AS total_trans, CASE WHEN COUNT(*) > 400 THEN 'mas de 400' ELS...	100 row(s) returned